

HOW TO FIND VOLUME OF REVOLUTION IN 5 EASY STEPS

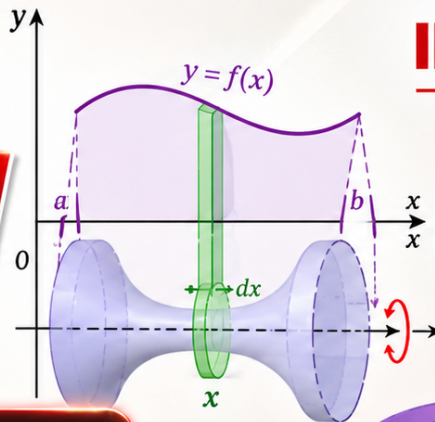
FOR **AAHL** & **AIHL**



IBMATH
PUNE

VOLUME

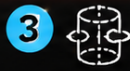
$$V = \pi \int_a^b [f(x)]^2 dx$$



1
IDENTIFY
THE REGION



2
CHOOSE THE
AXIS OF REVOLUTION



3
CHOOSE THE
METHOD
(Discs / Washers / Shells)



4
SET UP THE
INTEGRAL



5
EVALUATE
THE INTEGRAL



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AAHL

SL 5.5

Content	Guidance, clarification and syllabus links
Introduction to integration as anti-differentiation of functions of the form $f(x) = ax^n + bx^{n-1} + \dots$, where $n \in \mathbb{Z}$, $n \neq -1$	Students should be aware of the link between anti-derivatives, definite integrals and area.
Anti-differentiation with a boundary condition to determine the constant term.	Example: If $\frac{dy}{dx} = 3x^2 + x$ and $y = 10$ when $x = 1$, then $y = x^3 + \frac{1}{2}x^2 + 8.5$.
Definite integrals using technology. Area of a region enclosed by a curve $y = f(x)$ and the x -axis, where $f(x) > 0$.	Students are expected to first write a correct expression before calculating the area, for example $\int_2^6 (3x^2 + 4)dx$. The use of dynamic geometry or graphing software is encouraged in the development of this concept.

SL 5.11

Content	Guidance, clarification and syllabus links
Definite integrals, including analytical approach.	$\int_a^b g'(x)dx = g(b) - g(a)$. The value of some definite integrals can only be found using technology. Link to: definite integrals using technology (SL5.5).

Content	Guidance, clarification and syllabus links
Areas of a region enclosed by a curve $y = f(x)$ and the x -axis, where $f(x)$ can be positive or negative, <u>without the use of technology.</u> <u>Areas between curves.</u>	Students are expected to first write a correct expression before calculating the area. Technology may be used to enhance understanding of the relationship between integrals and areas.

AHL 5.17

Content	Guidance, clarification and syllabus links
Area of the region enclosed by a curve and the y -axis in a given interval. Volumes of revolution about the x -axis or y -axis.	

AIHL

AHL 5.12

Content	Guidance, clarification and syllabus links
Area of the region enclosed by a curve and the x or y -axes in a given interval.	Including negative integrals.
Volumes of revolution about the x -axis or y -axis.	$V = \int_a^b \pi y^2 dx$ or $V = \int_a^b \pi x^2 dy$



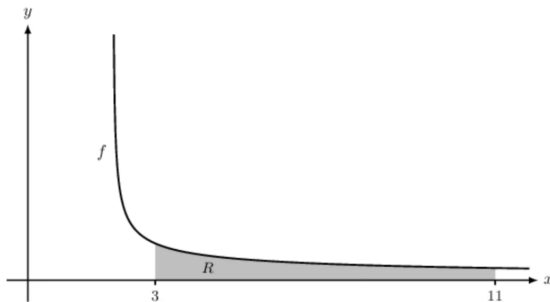
Question (1):

[Maximum mark: 5]

Let $f(x) = \frac{1}{\sqrt{3x-6}}$, for $x > 2$.

(a) Find $\int (f(x))^2 dx$. [2]

Part of the graph of f is shown in the following diagram.



The shaded region R is enclosed by the graph of f , the x -axis, and the lines $x = 3$ and $x = 11$.

(b) Find the exact volume of the solid formed when R is rotated 360° about the x -axis. [3]

$$(a) \int (f(x))^2 dx = \int \frac{1}{3x-6} dx = \frac{1}{3} \ln |3x-6| + C$$

(b) Volume of revolution is

$$v = \pi \int_3^{11} (f(x))^2 dx = \pi \int_3^{11} \frac{1}{3x-6} dx = \pi \left[\frac{1}{3} \ln |3x-6| \right]_3^{11}$$





$$\therefore V = \frac{\pi}{3} \{(\ln(3(11) - 6)) - \ln(3(3) - 6)\}$$

$$= \frac{\pi}{3} \{\ln 27 - \ln 3\} = \frac{\pi}{3} \ln \left(\frac{27}{3} \right) = \frac{\pi}{3} \ln(9) \text{ unit}^3$$



Question (2):

[Maximum mark: 6]

The graph of $y = f(x)$ is obtained from the graph of $y = \ln x$ by a horizontal translation 3 units in the negative x direction and a vertical translation $\ln 6$ units in the positive y direction.

- (a) Find $f(x)$. [2]
- (b) The region bounded by the graph of $y = f(x)$ and the two axes is rotated through 2π radians about the y -axis. Find the volume generated. [4]

$$(a) \quad f(x) = \ln(x + 3) + \ln 6 = \ln(6(x + 3)) = \ln |6x + 18|$$

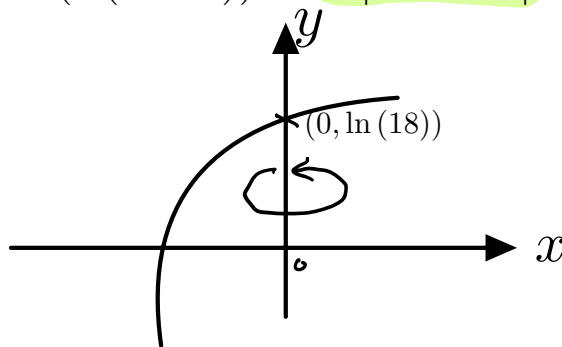
$$(b) \quad y = \ln |6x + 18|$$

$$\therefore e^y = 6x + 18$$

$$\therefore x = \frac{1}{6}(e^y - 18)$$

$$\therefore x^2 = \frac{1}{36}(e^y - 18)^2$$

$$\text{Volume about Y-axis} = \pi \int x^2 dy$$





$$\text{Volume about y-axis} = \pi \int_0^{\ln 18} x^2 dy$$

$$= \pi \int_0^{\ln 18} \frac{1}{36} (e^y - 18)^2 dy$$

$$\text{Volume about y-axis} \approx 42.4 \text{ units}^3 \quad (\text{by using G.D.C.})$$



AAHL & AIHL

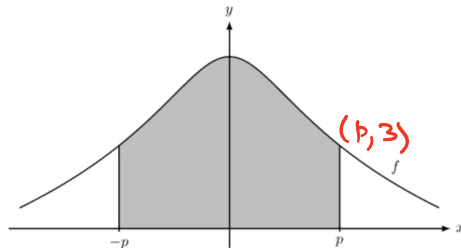
[Maximum mark: 5]

Let $f(x) = 9 - 2 \ln(x^2 + 4)$, for $x \in \mathbb{R}$. The graph of f passes through the point $(p, 3)$, where $p > 0$.

(a) Find the value of p .

[2]

The following diagram shows part of the graph of f .



The region enclosed by the graph of f , the x -axis and the lines $x = -p$ and $x = p$ is rotated 360° about the x -axis.

(b) Find the volume of the solid formed.

[3]

(a) f passes through $(p, 3)$

$$\therefore f(p) = 3$$

$$\therefore 3 = 9 - 2 \ln(p^2 + 4)$$

$$\therefore p = 4.01$$

(by using G.D.C.)



$$\therefore \text{Volume of revolution is } = \pi \int_{-4.01}^{4.01} y^2 dx$$

$$= \pi \int_{-4.01}^{4.01} (9 - 2 \ln(x^2 + 4))^2 dx$$

$$= 605 \text{ units}^3 \quad (\text{by using G.D.C.})$$

(604.7049...)



Practice questions PDF link is given in the description of this video.



1/09/2026
at

10 p.m.
Live



HOW TO SOLVE **HOMOGENEOUS** DIFFERENTIAL EQUATION IN **7** EASY STEPS


$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

Step-by-Step Method
Made Simple!

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 Perfect for **IB AAHL** Students

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